Series circuits

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use current continuity and voltage drops in a single-path circuit.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 4.*
5 J of electric potential-energy difference per coulomb.

02. Worked example: explain the model
Same 2 A through both; larger resistor has larger voltage drop.

03. Read and interpret the model
All current stops in the ideal single loop.

04. Apply the physics
R_{total}=6 Ω; I=1 A.

05. Explain your reasoning
Steady series current must be equal; unequal currents would change charge stored at the connection.

06. Review: connect an earlier idea *Review: Unit 6, Day 13.*
25° from the normal, on the other side.

07. Exit check
6 V.

08. STRETCH: test the model
Example 4+6 ohms: 1 A, drops 4+6 V; 2+8 ohms also works.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 4.*
21 J and 35 J; total 56 J.

02. Worked example: explain the model
Same 2 A through both; larger resistor has larger voltage drop.

03. Read and interpret the model
All current stops in the ideal single loop.

04. Apply the physics
30 Ω; 0.40 A; 4 V and 8 V.

05. Explain your reasoning
Steady series current must be equal; unequal currents would change charge stored at the connection.

06. Review: connect an earlier idea *Review: Unit 6, Day 13.*
Image moves 0.4 m toward the mirror from behind it. Separation decreases by 0.8 m.

07. Exit check
New resistance 3 R versus 2 R; current becomes 2/3 of the original.

08. STRETCH: test the model
Total 10 ohms, R 5. Powers I^2R: 8, 12, 20 W; sum 40 W.